

Test Project

Module C Functionality | Day 2 A.M.

Mobile Applications Development

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Introduction

WaveRider Co. is an emerging technology company dedicated to bringing the thrill of ocean sports and coastal culture to a global digital audience. The company plans to develop a series of mobile applications designed to help people experience the excitement of ocean sports, learn about marine life, and promote coastal tourism and healthy outdoor lifestyles. The GoSurfing app is the first in this planned series, targeting casual mobile gamers and ocean-sport enthusiasts alike.

The app is designed as an engaging, fast-paced surfing game that simulates the unpredictable nature of ocean waves. Players take on the role of a surfer navigating through increasingly challenging wave conditions, collecting shells, avoiding obstacles, and competing for the highest rankings. The game is intended to be intuitive enough for first-time players while offering enough depth and difficulty scaling to retain experienced users.

Today, the Product Manager has provided you with product wireframes for a phone game app about sea surfing. These wireframes outline the core structure, user flow, and visual layout of the application across all its pages — including the Home page, Game page, Rankings page, and Setting page. As a Mobile Application Developer, your task is to develop the corresponding app based on the given wireframes according to the specific development needs outlined in this document.

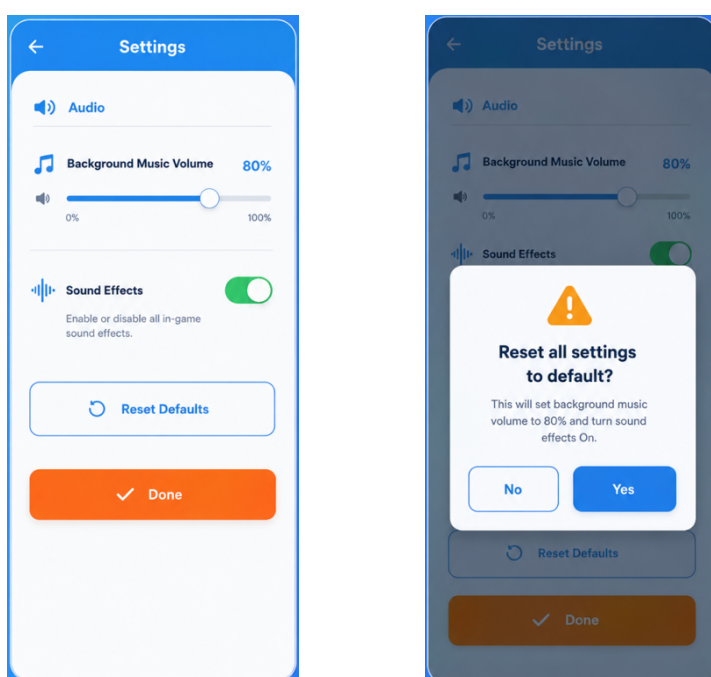
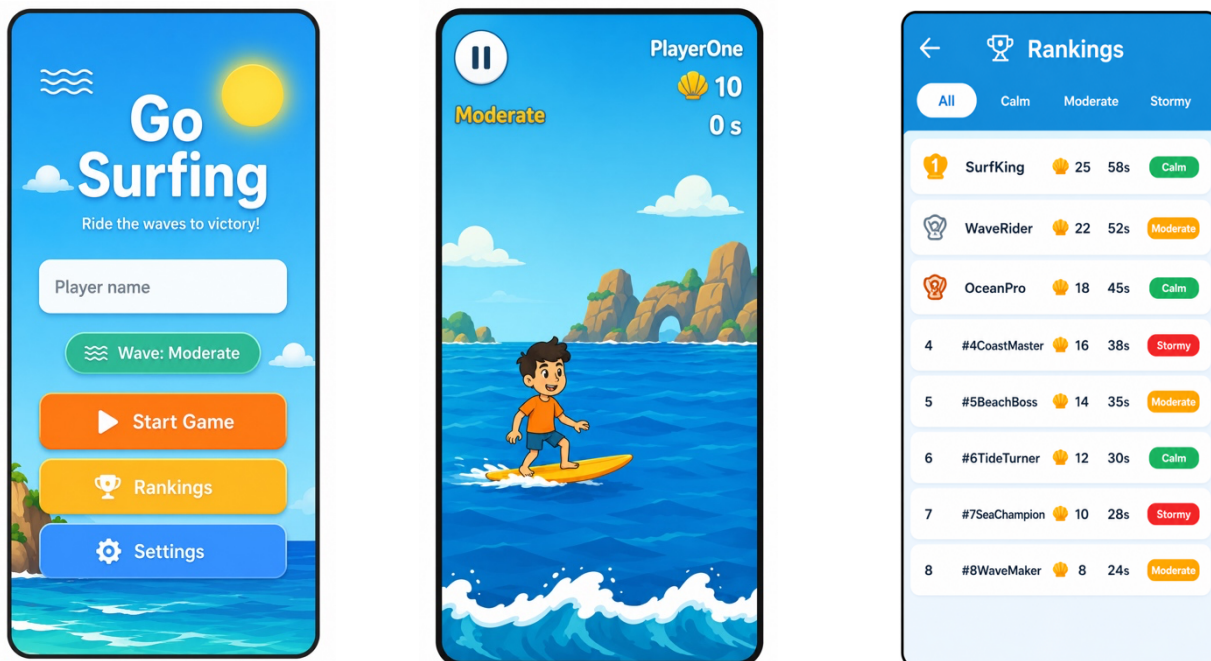
The application must be built for the Android platform and tested on an emulator. All gameplay mechanics, persistent data handling, audio behaviour, and UI interactions described in this document must be implemented fully and accurately. Functionality takes priority over visual polish — the interface does not need to match the wireframes pixel-perfectly, but all described behaviours must work correctly and reliably.

Description of Project and Tasks

You do not need to consider the program interface's aesthetics and the elements' position. The work can be inconsistent with the given wireframes, but you need to consider the ease of use.

NAME	COMPETITION TIME	DEVICES
Functionality (phone)	3 hours	Emulator

Below is an overview of some of the wireframes. Please check the **media-files** for details.



General Demands

- Finish the function of this application according to the demands;
- The application must remain in portrait orientation at all times, including all page transitions and interactions;
- The application must display the application name **"Go Surfing"** in the operating system;
- The application must display an application icon in the operating system. You must use **"go_surfing.png"** from the media-files as the app icon;
- All persistent data — including rankings and settings — must be fully retained after the application is closed and reopened;
- All sound effects and background music must play without noticeable delay at the moment they are triggered;
- The application must not crash or freeze during any interaction or state transition described in this document;
- All text displayed to the user must match exactly the wording specified in this document, including capitalisation and punctuation.

Pages Demands

Home Page

Elements include:

- Background image (ocean scene);
- **"Go Surfing"** title text;
- Player Name Input area:
 - Text input box;
 - Hint text "Player name" displayed inside the input;
- **"Start Game"** button;
- **"Rankings"** button;
- **"Setting"** button;
- Wave condition display area, showing one of three states (randomly assigned each session):
 - **"Calm"** — displayed in green;
 - **"Moderate"** — displayed in orange;
 - **"Stormy"** — displayed in red.

Functional requirements:

- Click the **"Setting"** button to navigate to the Setting page;
- Click the **"Start Game"** button:
 - If the player name input box is empty or contains only whitespace characters, a prompt box must pop up with the message "Invalid";
 - Leading and trailing whitespace in the player name must be trimmed before it is used or displayed anywhere in the game;
 - The player name input must accept a maximum of 20 characters; input beyond this limit must be prevented — the field must not accept further characters;
 - If a valid player name is entered, navigate to the Game page;
 - The wave condition displayed on the Home page is passed into the game session and affects gameplay as defined in the Game page requirements;
- Click the **"Rankings"** button to navigate to the Rankings page;
- The wave condition is randomly assigned each time the app is launched or each time the Home page is returned to after a completed game session;
- The player name entered must persist if the user navigates to Settings and returns to the Home page — re-entry must not be required.



Game Page (Default / Jump / Pause / Wipeout)

Elements include:

- Background image (ocean scene with sky, distant cliffs, and sea) — fixed, does not scroll;
- Surfer character on a surfboard, rendered using one of three sprite images: **Surfer Steady.png** (default/standing), **Duck Surfer.png** (duck mode), or **Splash Surfer.png** (wipeout);
- Animated white foamy wave line at the bottom of the screen;
- Pause button (||) positioned at the top left of the screen;
- Game status area on the top right:
 - Player Name text;
 - Shell icon followed by the shell quantity as a number;
 - Duration text in the format "X s" (e.g. "0 s", "15 s", "120 s");
- Wave condition indicator text displayed below the pause button on the top left, using the same colour coding as the Home page;
- **"Tilt Left"** button positioned at the bottom left of the screen, above the wave line;
- **"Tilt Right"** button positioned at the bottom right of the screen, above the wave line;
- **"Duck"** button positioned at the mid left of the screen, vertically centred between the wave line and the top of the screen.



Functional requirements:

Game Initialization

- When the game starts:
 - The default number of shells is 10;
 - The default duration displayed is "0 s";
 - Background music "bgm.mp3" begins playing in a seamless loop immediately;
 - The wave condition indicator on the game screen matches the condition assigned on the Home page.

Background and Environment

- The background image (ocean scene with sky, distant cliffs, and sea) remains fixed at all times — it does not scroll or move during gameplay;
- The wave line at the bottom of the screen is the only animated background element:
 - Animates continuously with a smooth, rhythmic up-and-down bobbing motion throughout all active gameplay;
 - Swiping upward anywhere on the screen immediately increases the wave line animation speed for 2 seconds, then smoothly returns to the original animation speed;
 - The upward swipe speed boost has a 3-second cooldown — further swipes during the cooldown must be ignored;
 - Obstacles and shells move across the screen from right to left at a speed determined by the wave condition and difficulty scaling — the background image itself does not move.

Surfer Behavior

- The surfer character:
 - Remains horizontally centred on screen at all times;
 - Sits on the wave line and bobs up and down continuously in sync with the wave line animation;
- Tapping any position on screen (excluding the pause button, tilt buttons, and duck button):
 - Makes the surfer perform a jump animation — the surfer leaps upward off the wave line and lands back after approximately 0.8 seconds;
 - Plays sound effect "jump.wav" immediately upon tap;
 - The jump animation lasts exactly 0.8 seconds, after which the surfer automatically returns to the wave line;
 - Tapping again during an active jump resets the 0.8-second jump timer from the beginning;
 - While jumping, low obstacles (rocks) pass safely underneath the surfer without triggering a collision;
 - While jumping, high obstacles (seagulls) still cause a Wipeout collision — seagulls are positioned at jump height and will collide with the surfer mid-air;
 - While jumping, the surfer cannot collect shells — collection only occurs when the surfer is on the wave line in the standing position.

Surfer Animation States

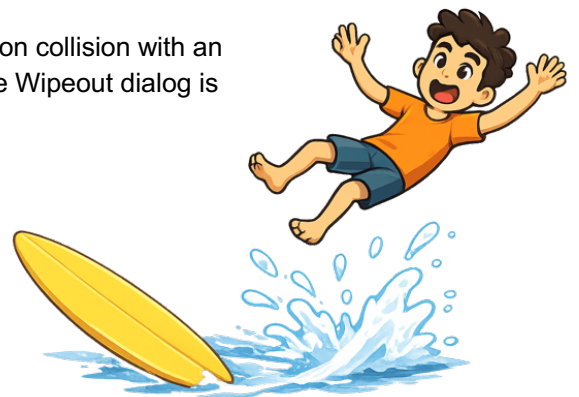
- **Standing / Default** — **Surfer Steady.png**; displayed whenever the surfer is on the wave line in normal gameplay or during tilt control;



- **Duck** — **Duck Surfer.png**; displayed only while the "**Duck**" button is held; the surfer crouches low to avoid seagulls;



- **Wipeout** — **Splash Surfer.png**; displayed immediately upon collision with an obstacle (when Shield Mode is not active); persists until the Wipeout dialog is dismissed.



Pause and Resume

- Clicking the pause button (||):
 - The surfer cannot be tapped or operated;
 - All wave animation, obstacle movement, and shell movement freeze immediately;
 - The duration timer stops counting;
 - The text "Game suspended..." is displayed at the centre of the screen;
 - The pause button icon switches to a play button (▶);
 - Background music stops immediately;
 - The Tilt Left, Tilt Right, and **Duck** buttons are disabled while paused.
- Clicking the play button (▶):
 - All animations, obstacle movement, shell movement, and the duration timer resume from exactly where they were paused;
 - The play button icon switches back to the pause button (||);
 - Background music resumes from the exact point where it stopped;
 - The Tilt Left, Tilt Right, and **Duck** buttons are re-enabled.

Wave Tilt Control (On-Screen Buttons)

- Two visible on-screen tilt control buttons are displayed during gameplay:
 - **"Tilt Left"** button positioned at the bottom left of the screen, above the wave line;
 - **"Tilt Right"** button positioned at the bottom right of the screen, above the wave line;
- Both buttons must be visible and tappable at all times during active gameplay, except when the game is paused.
- Tapping and holding the **"Tilt Left"** button:
 - The wave line's angle gradually decreases (flattens toward horizontal), animating smoothly down to 0° (fully flat);
 - All movement speeds (wave animation, obstacles, shells) decrease proportionally as the angle flattens;
 - The background music volume decreases proportionally as the angle flattens;
 - When the wave reaches 0° (fully flat), all movement stops completely and music volume reaches zero;
 - The surfer cannot jump or be interacted with while the wave is completely flat;
 - The surfer sprite remains **Surfer Steady.png** — tilt buttons do **not** trigger the Duck animation.
- Tapping and holding the **"Tilt Right"** button:
 - The wave line's angle gradually increases back toward the default inclined position, up to a maximum of 30°;
 - Movement speeds and music volume return to normal proportionally as the angle recovers;
 - Full surfer interaction is restored when the wave returns to its default inclined angle.
- Releasing either button stops the tilt change — the wave holds at whatever angle it reached when the button was released.

Duck Control (On-Screen Button)

- A **"Duck"** button is displayed at the mid left of the screen, vertically centred between the wave line and the top of the screen, at all times during active gameplay, except when the game is paused.
- Tapping and holding the **"Duck"** button:
 - The surfer immediately switches to the **Duck Surfer.png** sprite — the surfer crouches low on the board;

- While ducking, high obstacles (seagulls) pass safely overhead — no collision is triggered;
- While ducking, low obstacles (rocks) still cause a Wipeout collision;
- While ducking, the surfer cannot collect shells — collection only occurs in the standing position;
- The surfer cannot jump while the Duck button is held;
- Duck mode has no cooldown — it can be activated at any time during active gameplay.
- Releasing the **"Duck"** button:
 - The surfer immediately returns to the **Surfer Steady.png** sprite and resumes normal standing behaviour.

Quit Confirmation

- When the player swipes from right to left anywhere on the screen:
 - The game pauses immediately (same behaviour as pressing the pause button);
 - A confirmation dialog appears with the message **"The game is in progress. Are you sure to quit?"** and two buttons: **"Yes"** and **"No"**;
 - Selecting **"Yes"** stops the game, discards the current session (result is **NOT** saved to Rankings), and navigates to the Home page;
 - Selecting **"No"** dismisses the dialog and resumes the game as if the play button was pressed.

Duration and Scoring

- The duration counter:
 - Starts at 0 when the game begins;
 - Increases by 1 every second while the game is active;
 - Pauses when the game is paused (pause button pressed or quit confirmation dialog shown);
 - Resumes counting when the game is resumed;
 - Displays in the format "X s" at all times.
- The shell count displayed in the status area updates immediately whenever a shell is collected or consumed — there must be no visible delay.

Difficulty Scaling

- As the game duration increases, the game difficulty scales automatically:
 - Every 15 seconds, the obstacle and shell movement speed increases by 10% of the original base speed for the current wave condition;
 - Every 20 seconds, the obstacle spawn rate increases by one additional obstacle per spawn cycle;
 - The maximum movement speed is capped at 3× the original base speed, regardless of duration or wave condition;
 - "Stormy" will reach the cap faster than "Calm";
 - Speed scaling affects obstacle movement and shell movement equally — the background image remains fixed throughout.

Shield Mode

- Long-pressing anywhere on the screen (excluding the pause button, tilt buttons, and duck button) activates Shield Mode:
 - Shield Mode is activated only if the current shell count is greater than 0;
 - Upon activation, 1 shell is immediately deducted from the shell count;
 - The text "Shield Mode" appears at the centre of the screen for the entire duration of the mode;

- The surfer passes through all obstacles without any collision during Shield Mode;
- Every additional 1 second of continued long-press deducts 1 more shell from the count;
- Shield Mode is canceled immediately when: (i) the player releases their finger; or (ii) the shell count reaches 0;
- If the shell count is already 0 when a long-press is initiated, Shield Mode must not activate — no visual change or deduction occurs.

Combo System

- A combo counter tracks consecutive shell collections:
 - Each shell collected without hitting an obstacle increments the combo count by 1;
 - When the combo count reaches **3**: the text "**Combo x3!**" is displayed on screen for 1.5 seconds and "combo.wav" is played;
 - After reaching x3, the combo counter resets to 0 and begins counting again from the next shell collected;
 - Hitting any obstacle resets the combo counter to 0;
 - The combo counter number is not displayed on screen at all times — only the milestone text message is shown.

Game Over (Wipeout)

- When the surfer collides with an obstacle and Shield Mode is not active:
 - The surfer sprite immediately changes to **Splash Surfer.png** upon collision;
 - All movement, animation, and the duration timer stop immediately;
 - The game plays "wipeout.wav" from the media-files;
 - The current game result (player name, shell count, duration, wave condition) is automatically and immediately saved to the Rankings persistent storage;
- A pop-up dialog appears over the frozen game screen, containing:
 - The text "Wipeout!" in large text at the top of the dialog;
 - Player name label displaying the correct trimmed name from the Home page input;
 - Shell icon with the correct final shell count;
 - Duration displaying the correct final duration in "X s" format;
 - Wave condition label showing the current session's wave condition with its corresponding colour (green / orange / red);
 - **"Restart"** button — tapping this restarts the game with the same player name, a newly randomised wave condition, shells reset to 10, and duration reset to 0;
 - **"Go To Rankings"** button — tapping this navigates to the Rankings page with the just-completed record automatically highlighted.

Game Page (Obstacle / Shell)

Elements include:

- Low obstacles — rocks on the wave surface;
- High obstacles — low-flying seagulls above the wave line;
- Regular shells — collectible items floating on the wave;
- Golden shells — special collectible items, visually distinct with a golden colour and larger size.

Functional requirements:

Obstacles

- Low obstacles (rocks):
 - Appear from the right edge of the screen and move left at the current game speed;
 - Are positioned on the wave line at the surfer's standing body level;
 - Collide with the surfer when the surfer is in the standing position on the wave line;
 - Are safely avoided by the surfer performing a jump — the surfer leaps over the rock as it passes beneath;
 - Exit the screen at the left edge without any effect if not collided with.
- High obstacles (seagulls):
 - Appear from the right edge of the screen and move left at the current game speed;
 - Are positioned above the wave line, at approximately the surfer's jump height;
 - Collide with the surfer when the surfer is mid-jump;
 - Are safely avoided by the surfer remaining on the wave line in standing position, **or by activating Duck mode** — the seagull passes overhead while the surfer stays grounded or crouches;
 - Seagulls are visually distinguishable from rocks — a different sprite or image must be used.
- Obstacle spawning rules:
 - A minimum gap of 1.5 seconds of travel time must exist between any two consecutive obstacles;
 - A rock and a seagull must never appear at the same horizontal position simultaneously — a player must never be forced into a situation where jumping causes a seagull hit and standing causes a rock hit at the same moment;
 - Spawn frequency is governed by the wave condition: "Calm" = low, "Moderate" = medium, "Stormy" = high;
 - Spawn frequency also increases over time as per the difficulty scaling rule;
 - Obstacle type (rock or seagull) is determined randomly at spawn time, with roughly equal probability.

Shells

- Regular shells:
 - Appear from the right edge of the screen and move left at the current game speed;
 - Are positioned on the wave line at the same height as the surfer in the standing position;
 - When the surfer (in standing position) passes through a shell, the shell disappears immediately;
 - "shell.wav" plays upon collection;
 - The shell count in the status area increases by 1 immediately.
- Shells cannot spawn at the same horizontal position as an obstacle — a minimum gap of 0.5 seconds of travel time must be maintained;
- Golden shells:
 - Appear randomly with a probability of approximately 1 in every 8 shell spawns;
 - Are visually distinct from regular shells — golden/yellow colour and a slightly larger size;
 - When collected, "golden_shell.wav" is played;
 - The shell count increases by 3 (not 1) upon collection;
 - The text "+3" is displayed briefly at the collection point on screen for 1 second;
 - A golden shell counts as 1 (not 3) for the purposes of the combo counter.
- The surfer cannot collect any shell (regular or golden) while mid-jump or while in Duck mode — collection only occurs when the surfer is on the wave line in the standing position.

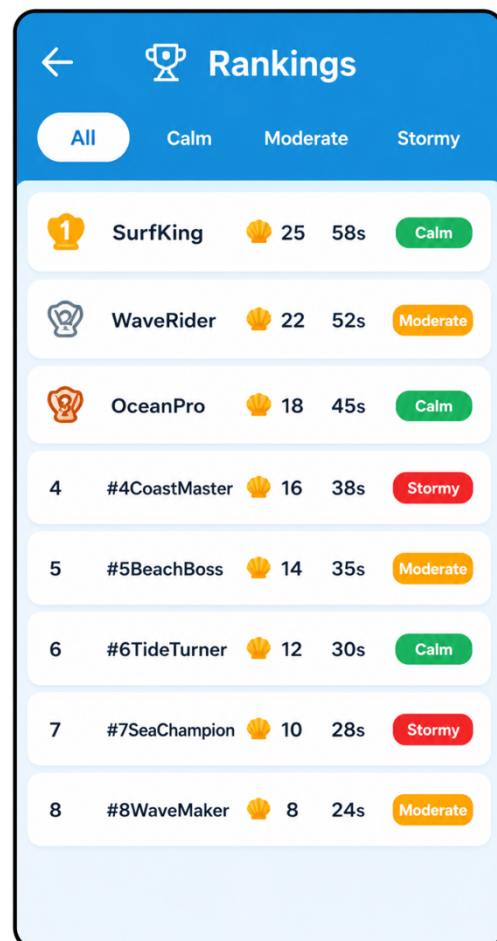
Rankings Page

Elements include:

- **"Rankings"** title text;
- Back button;
- Filter tab bar with four tabs:
 - "All";
 - "Calm";
 - "Moderate";
 - "Stormy".
- List of records, each item containing:
 - Rank number;
 - Player name;
 - Shell count (with shell icon);
 - Duration in "X s" format;
 - Wave condition badge (colour-coded: green / orange / red).
- **"No Ranking"** placeholder text when no records exist (or no records exist for the selected filter).

Functional requirements:

- Records are arranged in descending order by duration (longest duration = rank 1);
- Tie-breaking rules:
 - If two records have equal duration, they are sorted by shell count in descending order;
 - If both duration and shell count are equal, they are sorted by time of entry, most recent first.
- Rank numbers are recalculated and reassigned dynamically after every add or delete operation;
- When navigating to this page from the Game Over dialog, the most recently ended game record is automatically highlighted with a distinct background colour — no user interaction is required;
- When navigating to this page from the Home page, no record is highlighted;
- Ranking data is persistently stored — it must survive app closure and full device restart;
- Filter tab behaviour:
 - Tapping "All" shows all records regardless of wave condition;
 - Tapping "Calm", "Moderate", or "Stormy" shows only records from that wave condition;
 - Rank numbers shown under a filtered tab reflect the position within that filtered subset only, not the global overall rank;
 - If no records exist for the selected filter tab, the text "No Ranking" is displayed;
 - The currently active filter tab must be visually distinguishable from inactive tabs.
- Long-pressing any record in the list:
 - A confirmation dialog appears with the message **"Delete this record?"** and two buttons: **"Yes"** and **"No"**;
 - Selecting **"Yes"** permanently deletes that record, immediately refreshes the list, and recalculates rank numbers;
 - Selecting **"No"** dismisses the dialog without any change.
- The list supports scrolling when the number of records exceeds the visible screen area;
- Click the back button to navigate to the Home page.



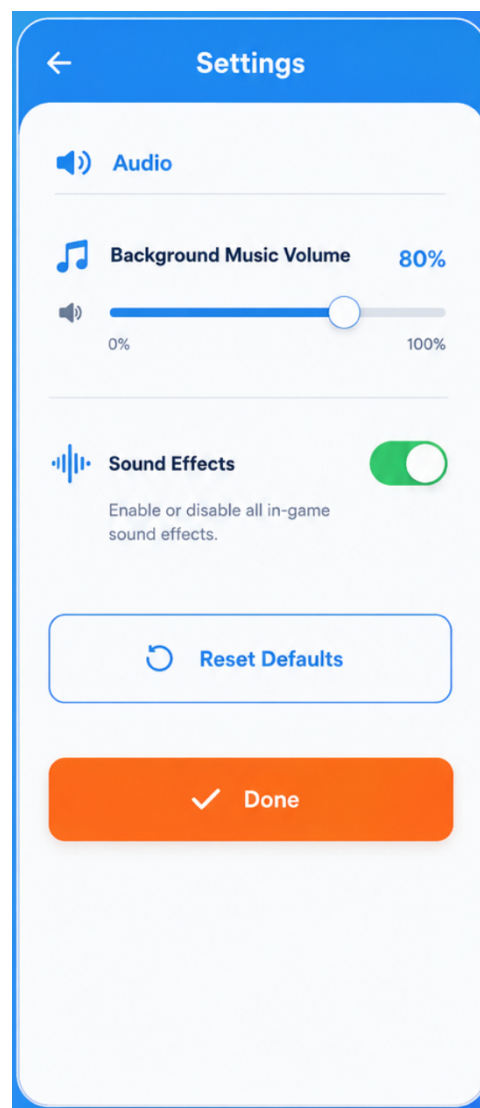
Setting Page

Elements include:

- Background music volume slider (range 0–100%, with percentage label);
- Sound effects toggle switch (On / Off);
- **"Reset Defaults"** button;
- **"Done"** button.

Functional requirements:

- The background music volume slider:
 - Controls the playback volume of "bgm.mp3" during gameplay;
 - Displays the current percentage value next to the slider (default "80%");
 - A value of 0% means the music plays silently — the music track continues running but is inaudible (it must not be paused or stopped);
 - The volume level set here takes effect from the start of the next game session;
 - The wave tilt control during gameplay overrides the slider volume temporarily — when the wave returns to its default angle via the Tilt Right button, the volume must return to the value set in Settings.
- The sound effects toggle:
 - When set to Off, all in-game sound effects are silenced: "shell.wav", "jump.wav", "wipeout.wav", "combo.wav", "golden_shell.wav";
 - When set to On, all sound effects play normally;
 - The toggle state must be visually clear — the On and Off states must have a distinct, clearly different appearance;
 - The background music ("bgm.mp3") is not affected by the sound effects toggle — it is controlled separately by the volume slider.
- Tapping **"Reset Defaults"**:
 - A confirmation dialog **"Reset all settings to default?"** with **"Yes"** and **"No"** must appear before applying any reset;
 - Selecting **"Yes"** applies the following default values: volume = **80%**, sound effects = **On**;
 - The slider position must update immediately in the UI to reflect the reset values;
 - Selecting **"No"** dismisses the dialog without any change.
- All settings are persistently saved — they must be retained even after the app is fully closed and reopened;
- Settings are saved automatically when the user taps **"Done"** — there is no separate save button;
- Click the **"Done"** button to navigate to the Home page.



Instructions to the Competitor

- Please create the project with the correct package name (Android) / Organization Identifier (iOS). The naming convention is CC.ws2026.XX.am;
- You should save the project in the folder XX_Module_C_AM;
- You should rename the generated apk (Android) / app (iOS) file as XX_Module_C_AM.apk (Android) / XX_Module_C_AM.app (iOS) and save it in the root of the XX_Module_A_AM folder;
- Push the entire XX_Module_C_AM folder to the provided remote Git repository before time is called.

Project Name	Go Surfing
Package Name Convention	CC.ws2026.XX.am
App Icon File	go_surfing.png
Folder Name	XX_Module_C_AM
APK / APP File Name	XX_Module_C_AM.apk / XX_Module_C_AM.app
Competition Time	3 hours
Marking Device	Emulator
Media Files Required	bgm.mp3, jump.wav, wipeout.wav, shell.wav, golden_shell.wav, combo.wav, go_surfing.png, Surfer Steady.png, Duck Surfer.png, Splash Surfer.png